Massive Scale Data Analytics using Spark

Retrospective Report

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1 Timeline Comparison

Week 1:

- **Planned approach:** Learning PySpark library and setting up the environment as cluster access would come at a later stage.
- **Actual approach:** Set up the environment, learnt some core Spark concepts and looked out for a Larger enough dataset to use for the later stages of the project.
- Largely this stage aligned well with what I had set out to do within the first week.

Week 2:

- **Planned approach:** Implement a local version of a Word Count on a relatively small dataset.
- **Actual approach:** No cluster access was given; tried working with smaller subsets of the larger Wikipedia dump I had found.
- At this stage I was not certain when the access to the Big Data cluster would come and had to learn as much about the dataset I had chosen to effectively process the data.

Week 3 - 4:

- **Planned approach:** Scaling the Word Count program to the larger dataset and testing it's performance.
- **Actual approach:** Decided to look into the reading of the XML dataset and running the cleaning regex code on the smaller subset which could be scaled on the cluster later.
- I did not have access to the Cluster at this stage so had to move the Word Count program scaling to a later week.

Week 5 - 6:

- **Planned approach:** Implement the Distributed Grep implementation and scaling.
- **Actual approach:** Implemented the Word Count and scaled it once cluster access was available.
- Word Count moved to this stage due to cluster access being slightly later than planned.

Week 7 - 8:

- **Planned approach:** Further implementation of the Distributed Grep program.
- **Actual approach:** Implemented the Distributed Grep application using the same cleaning and pre-processing flow as Word Count.
- Mostly aligned, but later than planned due to earlier shifts made to the schedule.

Week 9 - 10:

- **Planned approach:** Record performance analysis for the programs on the Big Data cluster while changing configurations.
- **Actual approach:** Recorded performance analysis on the different spark configurations but also performed analysis on different dataset sizes.
- This stage stayed as planned apart from more tests having to be implemented on the different dataset sizes.

Week 11:

- **Planned approach:** Prepare the interim report on the early deliverables.
- **Actual approach:** The records of the experiments on the cluster were gathered on excel and displayed as plots using Matplotlib and then was able to describe any findings in the interim report.
- Plan did not account for any data visualisation.

2 Term 2 Plans

- **Week 1:** Scope and setup - confirm the datasets being used and the data mining algorithm to allow for time to prepare. Discuss methods of testing distributed programs.
- **Week 2 - 4:** Implement a statistical analysis on a simple data set. Show frequency distribution and different order statistics. Then visualise the shape of distribution.
- **Week 5:** Conduct performance testing and capture runtime and throughput, produce graphs and brief configuration tuning notes.
- **Week 6 - 9:** Implement the advanced data mining algorithm selected and apply it on a realistic dataset, discovering any interesting features.
- **Week 10:** Performance testing and recommendations on the advanced data mining implementation, produce graphs and brief configuration tuning notes.
- **Week 11:** Polish the report and the poster as well as any final deliverables.

3 Reflection

What went well during term 1 of the project were that the implementations of the Word Count and Distribute Grep were completed and successfully ran on the Big Data cluster. Using the smaller datasets from the beginning gave me hands on experience with the data I would be using in the large 100 GB Wikipedia XML data dump, which allowed for a much more streamlined process when I eventually started to attempt the up-scaling process. I was able to run multiple different tests, which lead to a variety of results that correlated in the way I expected due to the background research. I was able to use tmux to conduct my performance tests, which saved time as I was able to work on other things when the tests were being executed over the cluster. Having access to the CIM help desk assisted in resolving server related issues and provided clarity on any problems I would run into.

Some things that did not go well were, the cluster being quite unstable and causing access issues which reduced the efficiency of the number of performance tests. I found out by sending a ticket to the CIM help desk that the server was having hardware issues and was down for maintenance that limited the access I had and due to that some of the limitations of tmux became apparent as the performance tests I conducted had been wiped and would need to be re-run. I could have implemented a log file that generated at the end of each performance test but since it was quite late in the timeline I decided to implement that in term 2 across the other deliverables.

I realised quite late that I could have increase the number of partitions which itself would improve the performance and provide a great insight into the programs and clusters performance with more partitions increasing the parallelisable aspect of the stages across the cluster. This will be another key implementation in the performance tests conducted on the programs created in term 2.

For the future implementations, I would like to create some kind of testing on the distributed programs creating a more robust environment for the code, as well as automate the collection of the runtime data for each job lowering the risk of losing all the progress made from a computation or job in progress being interrupted by the server going down. I also intend to use visualisation tool Power Bi as it's a technology used industry wide and getting experience with the tool will allow for better visual presentations of the performance and statistical data I gather.

4 Risk Assessment

4.1 Technical risks

A risk I had identified was the risk for version and compatibility issues with AWS S3 or the cluster, with my mitigation being to have a list of versions and using HDFS as an alternative. Through the course of Term 1, I was able to mitigate any risk when I got on the cluster by contacting the CIM help desk to inform me of the versions of each Python, Hadoop and Spark being used of the Big Data cluster, which i stored within my dairy file and can also carry into Term 2. I was not able to use AWS S3 so had to use HDFS instead as a way to store the large files, I will look to try and use if for the next Term with access from the supervisor.

4.2 Limited access to resources

Another key risk was the limited access to the Cluster due to downtime or maintenance delaying any experiments; my mitigation was to schedule a time to access the cluster and have Spark on my local machine to run any smaller tests. Unfortunately, this was a major issue during the Term as there were periods where my tests failed due the server was completely inaccessible for entire days on which I had planned to conduct some performance tests of the implementations. Additionally, not having server access from an earlier stage limited the overall time I had to learn the system to an advanced level. A possible way of mitigating this risk in Term 2 would be to run the performance tests at an earlier stage, potentially after each implementation has been made. Another mitigation not entirely in my hands, is the replacing of some faulty hardware during maintenance, which should help keep the server up without as many issues.

4.3 Hardware failure

The risk of losing the code and performance test data were of high importance and were planned for by using version control system, GitLab to have backups of the files and to ensure the safeguarding of the codebase. I mitigated for the code with fair effectiveness, although the experimental progress was still at risk due to the server instability as a performance test if cut in the middle of its run would require a re-run. To mitigate this risk further, after each test run, storing the output data relevant for the performance testing on the HDFS or cloud immediately would be the way forward.

4.4 Time management

Poor time management was a risk due to underestimating the difficulty of the tasks; the mitigation for this was to allocate adequate time within the schedule for processes that may take longer. During Term 1, the work assigned by the deliverables was completed with the implementation and performance testing for each problem. Due to server access issues this forced a re-ordering of the timeline.

4.5 Data availability

I figured another risk would be that the dataset may not be large enough for the project and to mitigate this I could use Kaggle and other suitable datasets then upscale.

In term 1, this issue did not occur as I was able to find a site of Wikipedia dumps of different sizes and could scale the amount I required. Although for Term 2, I will be dealing with numeric datasets but I will continue to work with smaller subsets as a validation process and to save time if any errors occur. Additionally, finding a suitable dataset for the advanced data mining will be a priority from the start.